Residual Surrogates for Nonlinear DSGE Estimation: An Auditable Target-Corrected Workflow

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Abstract

Repeated nonlinear solution can dominate likelihood-based estimation of dynamic stochastic general equilibrium models. We develop an auditable workflow in which a neural network learns the difference between extended-path trajectories and a parameter-specific first-order solution. Cheap surrogate gradients form Hamiltonian proposals; an endpoint Metropolis test preserves the density actually evaluated there. This distinction matters because correcting for a surrogate profile objective is not exact inference for the nonlinear model. On the complete 23-equation Galí model, preserved deterministic pilots solve 80 targets with maximum residual 9.92×10^{-9}, improve calibration-set error over the first-order model by 53.9 percent, and match Python and JAX evaluations within 1.71×10^{-13}. These results establish software feasibility, not posterior fidelity or speed. The paper therefore couples the method with frozen gates for stochastic-solution accuracy, independent parity, untouched test performance, posterior agreement, multichain sampling, and same-work timing.

1 Introduction

Likelihood-based estimation of a nonlinear DSGE model repeats a costly numerical experiment at every parameter proposal. A single evaluation may require a steady state, a local solution, nonlinear conditional trajectories, latent-shock inversion, and measurement-error evaluation. Posterior simulation then repeats this stack thousands of times. Approximation is attractive, but an unmeasured approximation can change the objective, the posterior geometry, and even the numerical support of the model.

This paper develops a residual-surrogate workflow designed around that risk. The method solves the complete nonlinear equations at designed parameter, state, and shock inputs, then learns the difference between those solutions and a parameter-specific first-order reduced-order model (ROM). The ROM supplies a disciplined local baseline; the network is responsible only for the remaining nonlinearity. During sampling, the cheap differentiable objective supplies the Hamiltonian force. A separately declared density is evaluated at each proposal endpoint. Standard Metropolis correction then preserves that declared density when the numerical proposal map is reversible and volume preserving.

The preserved Galí pilots establish that this software path composes on the complete 23-equation nonlinear model while only the estimated parameter vector is reduced. Across 80 deterministic extended-path targets, the largest recorded equation residual is 9.92×10^{-9}. Python and JAX evaluations of the same surrogate profile objective differ by at most 1.71×10^{-13} at the audited calibration. A calibration block shows a 53.9 percent mean reduction in state error relative to the ROM after per-output shrinkage is fitted on that same block. The last number is not an untouched

test result, the HMC budgets are not convergence evidence, and no preserved run yet establishes a nonlinear posterior speed advantage.

The paper makes three contributions. First, it separates three objects often conflated in surrogate estimation: nonlinear trajectories used as labels, a surrogate profile objective used for cheap differentiation, and the nonlinear likelihood required for a structural-posterior claim. Second, it integrates parameter-dependent DSGE solution, residual learning, bounded-parameter HMC, and explicit failure handling in a content-addressed workflow. Third, it turns publication claims into executable gates covering stochastic expectations, cross-language parity, untouched test accuracy, posterior fidelity, sampler diagnostics, and full computational accounting.

The paper does not claim that endpoint correction or neural solution methods are new by themselves. Its narrow claim is that nonlinear DSGE surrogates can be made auditable enough to determine when they are safe and useful for estimation. The rest of the paper defines the estimands, describes the numerical workflow, proves the target-invariance statement used by the sampler, and reports the preserved engineering evidence before stating the confirmatory design and stopping rules.

1.1 Relation to the literature

The estimation problem sits between nonlinear DSGE solution, neural approximation, and accelerated MCMC. Extended path provides the nonlinear trajectory construction [Fair and Taylor, 1983]; stochastic extended path replaces future-shock means with numerical integration over conditional expectations [Adjemian and Juillard, 2026]. Likelihood construction for nonlinear DSGE models remains model- and solution-dependent [Kollmann, 2017], which is why this paper treats the trajectory solver, latent-shock treatment, and statistical objective as separate validation targets.

Neural networks have been used to solve high-dimensional dynamic economic models [Maliar et al., 2021]. Here the network does not replace the economic model wholesale: it learns the full-minus-first-order discrepancy and is always benchmarked against the ROM. Surrogate-gradient HMC and neural-gradient HMC already establish the proposal-side idea [Zhang et al., 2017, Li et al., 2019]. Our focus is the less forgiving DSGE case in which the endpoint density may contain parameter-dependent solution failures and may differ from the density used to generate the force. The independent implementation benchmark uses MacroModelling.jl [Kockerols, 2023], while the JAX path builds on the differentiable state-space perspective of Childers et al. [2022].

2 Economic model and statistical objects

Let $\theta \in \Theta$ denote structural parameters, s_t predetermined states, x_t endogenous variables, and ε_t structural shocks. Write the nonlinear equilibrium system as

$$F(x_{t-1}, x_t, \mathbb{E}_t x_{t+1}, \varepsilon_t; \theta) = 0. \tag{1}$$

For a given parameter vector and shock design, a nonlinear extended-path solver produces a trajectory $x^F_{1:T}(\theta)$ from the full-order model (FOM). A first-order solution produces the cheaper trajectory $x^R_{1:T}(\theta)$. We fit a network r_ψ to the residual

$$\Delta x_t(\theta) = x^F_t(\theta) - x^R_t(\theta), \qquad \widehat{x}^F_t(\theta) = x^R_t(\theta) + r_\psi(z_t, \theta), \tag{2}$$

where z_t contains the states and shocks available at the prediction point. Parameter vectors, rather than individual rows, are assigned to data partitions. This prevents periods from the same parameter-specific trajectory from appearing on both sides of a split.

Table 1: Numerical objects and the claims they can support

Object	Construction	Valid interpretation
Nonlinear label x^F	Extended-path solution of the complete equilibrium system	Supervised target for trajectory approximation; not a probability density
Surrogate objective $\widetilde{L}_\psi$	ROM plus learned residual, with shocks profiled by (4)	A declared pseudo-posterior target when combined with a prior and transform Jacobian
Nonlinear likelihood L_F	Independently validated statistical likelihood with latent variables and Jacobians treated explicitly	Structural likelihood and posterior in (3)

The target nonlinear posterior for data $y_{1:T}$ is

$$\pi_F(\theta \mid y) \propto p(\theta) L_F(y_{1:T} \mid \theta), \tag{3}$$

where L_F denotes an independently validated nonlinear likelihood with a declared treatment of latent shocks and every required change-of-variables term. The present implementation instead defines a surrogate inversion objective $\widetilde{L}_\psi$ by replacing nonlinear trajectories with (2), optimizing latent shocks, and evaluating the shock and measurement penalties at that optimizer. Unless an integration, Laplace adjustment, or required change-of-variables term is supplied, this is a profile or pseudo-likelihood objective rather than a marginal likelihood. The two objects are therefore scientifically distinct. Agreement between Python and JAX evaluations of $\widetilde{L}_\psi$ is implementation parity; it is not evidence that $\widetilde{L}_\psi = L_F$.

More explicitly, let $h_\psi(s_t, \varepsilon_t, \theta)$ map a state and structural shock into the observables implied by the ROM plus residual surrogate. At each date, the implementation computes

$$\widehat{\varepsilon}_t(\theta) = \arg\min_{\varepsilon} \left\{ \|R^{-1/2}[y_t - h_\psi(s_t, \varepsilon, \theta)]\|_2^2 + \|Q^{-1/2}\varepsilon\|_2^2 \right\}, \tag{4}$$

using a damped Gauss–Newton iteration, and then sums the Gaussian penalties evaluated at $\widehat{\varepsilon}_t(\theta)$. Here R and Q are diagonal in the pilot. The state passed to date $t+1$ follows the configured ROM or surrogate replay. Equation (4) makes the missing operation clear: the latent shocks are optimized, not integrated out. It also clarifies why agreement between two implementations of this calculation is a software check rather than likelihood validation.

Table 1 records the three numerical objects used below. A trajectory label is not a density, and a profile objective is not a marginal likelihood. Keeping their names distinct prevents a correct MCMC kernel from being used to support the wrong inferential claim.

The benchmark is the complete nonlinear New Keynesian system distributed with the MacroModelling.jl examples [Kockerols, 2023] and associated with Galí [2015]. It contains 23 equations, 16 declared parameters, and three structural shocks. The reduced experiment estimates the Taylor-rule responses to inflation and output, ϕ_π and ϕ_y, while retaining every model equation. Reducing the parameter dimension makes a numerical posterior reference feasible; it does not replace the model with the familiar three-equation linearization.

3 Residual-surrogate construction

For training case i, let $u_i = (z_i, \theta_i)$ and $d_i = x_i^F - x_i^R$. Input and output location and scale are estimated only from the training partition. In standardized coordinates, the implemented data-fit

loss is

$$\widehat{\psi} = \arg\min_{\psi} \frac{1}{\sum_{i\in\mathcal{I}_{\mathrm{tr}}} w_i} \sum_{i\in\mathcal{I}_{\mathrm{tr}}} w_i \left\| \widetilde{d}_i - r_\psi(\widetilde{u}_i) \right\|_2^2, \tag{5}$$

with zero weight for masked failures. The optimizer is AdamW with gradient clipping. The network embeds states and shocks, conditions hidden activations on the full estimated parameter vector through feature-wise affine modulation, and uses residual blocks with SiLU activations. Deployment restores the original units and adds the predicted discrepancy to the ROM.

The implementation can fit an additional per-output shrinkage coefficient in $[0, 1]$ after network training. In the preserved pilots, those coefficients were fitted on the same parameter block used to report the resulting RMSE. That block is therefore a calibration set, not an untouched test set. We retain the values as engineering diagnostics but do not treat them as out-of-sample evidence. Publication launchers disable this calibration; a future use of shrinkage requires a distinct calibration partition and an untouched test partition.

3.1 Offline and online computations

The offline stage draws complete parameter vectors, solves a parameter-specific steady state and first-order ROM, and generates nonlinear extended-path targets conditional on designed state and shock paths. Each attempted target retains a case identifier, solver status, residual norm, and parameter-group identifier. Only the training partition determines feature scaling and network weights. Validation partitions are disjoint in complete parameter vectors, not merely in individual rows. The residual network and every preprocessing constant are then frozen and serialized with the model identity and source hashes.

The online stage transforms an unconstrained vector q to bounded structural parameters, adds the transform Jacobian to the declared log target, rebuilds the parameter-dependent objects required by the selected runtime, and evaluates either the profile objective or a nonlinear reference. A surrogate-gradient trajectory costs many cheap force evaluations but only one new target evaluation per transition. This accounting—rather than draws per second alone—is central when the endpoint target contains a nonlinear solution.

Extended path traces back to Fair and Taylor [1983]. Deterministic extended path sets future innovations to their conditional means and therefore does not capture Jensen effects in nonlinear expectations. Stochastic extended path approximates those conditional expectations over future shocks [Adjemian and Juillard, 2026]. This is why positive branching order, multiple nodes, and active variation in every shock are hard requirements for confirmatory evidence, even though order zero is useful for debugging deterministic nonlinearity.

4 Target-corrected surrogate trajectories

Let q be an unconstrained parameterization of θ, let p be Gaussian momentum with kinetic energy $K(p)$, and let $\pi_T(q)$ be the density the sampler is intended to preserve. A cheaper differentiable density $\pi_P(q)$ defines the force used in L leapfrog steps. Denote the resulting deterministic map by $(q', p') = \Phi_{\pi_P}(q, p)$. The acceptance probability is

$$\alpha\{(q,p),(q',p')\} = \min\left\{1, \exp\left[-H_T(q',p') + H_T(q,p)\right]\right\}, \quad H_T(q,p) = -\log \pi_T(q) + K(p). \tag{6}$$

Proposition 1 (Target invariance)**.** *Suppose Φ_{π_P} is reversible after momentum negation and volume preserving, and π_T is evaluated at both endpoints used in* (6)*. Then the Metropolis transition leaves π_T invariant, even when $\pi_P \neq \pi_T$.*

Proof. On the extended state $z = (q, p)$, the target density is proportional to $\exp\{-H_T(z)\}$. Compose the leapfrog map with momentum negation. By the assumptions, the resulting map is an involution with unit Jacobian determinant. The deterministic Metropolis ratio is therefore $\exp\{-H_T(z') + H_T(z)\}$, and (6) gives detailed balance on the extended state. Marginalizing the refreshed momentum leaves $\pi_T(q)$ invariant. The construction of Φ_{π_P} affects acceptance but not this invariance argument. □

One transition has four auditable steps. Draw fresh Gaussian momentum; apply L leapfrog steps using only $\nabla \log \pi_P$; reject immediately if the proposal or target is non-finite; otherwise apply (6) using π_T at the two endpoints. Step size and mass-matrix adaptation end with warmup. The frozen network, normalization constants, parameter transform, and target implementation are part of the recorded sampling state.

The proposition is the standard deterministic-proposal Metropolis argument for leapfrog HMC [Neal, 2011]. It places two practical restrictions on the implementation. First, the surrogate and every preprocessing transformation used in the force must be frozen during production sampling. Second, "exact" always modifies the declared target. If $\pi_T \propto p(\theta)\widetilde{L}_\psi$, the chain is exact for the surrogate-defined posterior, not for (3). Exact nonlinear posterior claims require endpoint evaluation of L_F itself.

For bounded parameters $\theta_j \in (a_j, b_j)$, the implementation samples $q_j \in \mathbb{R}$ and uses

$$\theta_j = a_j + (b_j - a_j)\,\mathrm{logit}^{-1}(q_j), \qquad \log J(q) = \sum_j \left[\log(b_j - a_j) + \log \sigma(q_j) + \log\{1 - \sigma(q_j)\}\right]. \quad (7)$$

Thus the endpoint target is defined in q coordinates and must include $\log J(q)$. Invalid steady states, indeterminate local solutions, failed nonlinear paths, or non-finite objective values must be rejected as points outside support. A large finite penalty is not equivalent: it creates an artificial plateau with positive density. The preserved pilot still uses such a sentinel and therefore cannot support an exact structural-posterior claim.

The sampler has three implementation-level regression checks. When proposal and target densities coincide, its draws and accept/reject decisions match the ordinary static-HMC implementation under the same key. Under a deliberately shifted and rescaled proposal force, a fixed-seed multi-chain test recovers the known moments of a standard-normal target after endpoint correction. Finally, non-finite candidate targets receive zero acceptance probability. These tests check kernel plumbing; they do not replace convergence and posterior-fidelity evidence for the DSGE application.

This distinction is especially relevant for differentiable state-space and DSGE estimation, where automatic differentiation improves computation but does not resolve approximation error in the target [Childers et al., 2022]. Related work has made nonlinear DSGE likelihoods more tractable [Kollmann, 2017]; our evaluation therefore treats accuracy against an independently computed nonlinear reference as a primary outcome.

5 Validation design

The confirmatory design has two tiers. Tier 1 estimates (ϕ_π, ϕ_y) and uses a dense posterior grid. Grid resolution is doubled until posterior moments and interval endpoints stabilize. This tier isolates numerical correctness from high-dimensional MCMC behavior. Tier 2 estimates a prespecified four-to-six parameter subset and measures scaling with four independently initialized chains. The paper will report Tier 2 only after Tier 1 passes.

Training, validation, and posterior-region designs are disjoint in parameter vectors, shock sequences, and trajectories. Validation includes regular prior-interior points, posterior-region points,

and a boundary-stress stratum. The last stratum is reported separately rather than deleted through a stability- margin screen. For stochastic extended path, evidence runs use positive branching order, multiple quadrature nodes, and nonzero coverage of every declared shock. Common case identifiers are evaluated by both the Python and MacroModelling.jl implementations.

For trajectory case i with d_x reported outputs, the primary approximation metric is the scale-adjusted root mean squared error

$$e_i^x = \left\{ \frac{1}{T d_x} \sum_{t=1}^{T} \left\| D_x^{-1} (\widehat{x}_{it}^F - x_{it}^F) \right\|_2^2 \right\}^{1/2}, \tag{8}$$

where D_x is fixed from the training partition. Paired differences $e_{i,\mathrm{ROM}}^x - e_{i,\mathrm{sur}}^x$ are summarized with a 95 percent interval over prespecified test cases. Reporting includes medians, upper-tail quantiles, and every output separately; an average cannot hide failure in one observable.

Likelihood geometry is compared after removing an arbitrary additive constant. For reference point θ_0, define

$$e^{\ell}(\theta) = \left[\log \widetilde{L}_{\psi}(\theta) - \log \widetilde{L}_{\psi}(\theta_0) \right] - \left[\log L_F(\theta) - \log L_F(\theta_0) \right]. \tag{9}$$

The grid analysis reports the distribution of e^{ℓ}, rank reversals near the mode, and normalized gradient discrepancies only where the reference gradient exceeds a frozen floor. Posterior means and interval endpoints are compared to their combined grid-discretization or Monte Carlo uncertainty, not to an arbitrary number of printed decimal places.

The primary accuracy outcomes are scaled equilibrium residuals; held-out, per-output trajectory errors; paired improvement over the ROM with uncertainty; centered log-likelihood error; rank reversals near the mode; and normalized gradient error away from zero-gradient cases. Posterior comparison reports differences relative to pooled Monte Carlo error, interval endpoint differences, and a distributional diagnostic. Sampling reports rank-normalized split $\widehat{R}$, bulk and tail effective sample sizes (ESS), ESS per gradient evaluation, and ESS per second.

Timing uses the same workload, precision, and hardware for both methods. Compilation, host callbacks, device transfers, and training are reported separately. Warm likelihood microbenchmarks use repeated synchronized calls; end-to-end comparisons use independent processes. The current Schur QME path invokes a sequential SciPy host callback, so a GPU label does not imply a GPU-native likelihood. The available GPU-native matrix-sign alternative must pass Galí-specific solution and likelihood parity before it enters a speed claim.

The primary decision is sequential. Experiment A first asks whether the two-parameter nonlinear reference and surrogate conclusions agree within the frozen numerical tolerances. A failure stops escalation to the higher-dimensional sampling experiment. Conditional on correctness, Experiment B asks whether the method improves effective samples per second after offline target generation, training, compilation, transfers, and failed solves are included. A method can therefore pass the correctness gate and fail the computational-value gate.

The protocol is frozen before confirmatory execution. It records the repository revision, dirty-worktree state, environment, command, seeds, hardware, precision, and content hashes for every evidence artifact. Numerical failures remain in attempted-case denominators and are stratified by prior-interior, posterior-region, and boundary-stress designs. Machine-readable evidence is the source of manuscript tables; hand-entered numerical cells are not permitted.

Table 2: Preserved full-model Galí engineering pilots

Run	Target runtime	$N_\theta \times T$	Max. residual	Calib. RMSE	Corrected	Accept.	Draws/s
Quick smoke	fixed-reference	4×4	7.21e-09	9.17e-05	No	1.000	0.378
Corrected resource	full-jax	4×4	7.21e-09	9.17e-05	Yes	1.000	0.0267
Default pilot	fixed-reference	8×10	9.92e-09	7.86e-06	No	1.000	8

Notes: All runs use the complete 23-equation nonlinear Galí source but estimate only two policy-rule parameters. N_θ is the number of parameter vectors and T is the observation horizon. Target construction used branching order zero and one node, so these are deterministic nonlinear extended-path pilots, not stochastic-expectations SEP validation. The reported calibration RMSE is computed on the parameter block also used to fit per-output residual-shrinkage factors; it is not an untouched test error. "Full-JAX" recomputes the parameter-dependent steady state and first-order ROM inside the surrogate profile inversion objective. It does not evaluate a full nonlinear SEP likelihood inside HMC. Correction therefore targets the declared surrogate objective, not an exact full-nonlinear posterior. The quick, corrected-resource, and default runs use respectively 1, 2, and 1 chains with 2, 2, and 50 post-warmup draws per chain. These runs establish execution and resource feasibility only.

6 Preserved engineering pilot

The pilot establishes three narrow facts. The parser, parameter-specific preflight, nonlinear equation solver, residual network, likelihood, and sampler compose on the full model. The order-zero extended-path targets satisfy a strict residual check in this small design. Finally, the Python and JAX implementations of the same surrogate profile inversion objective agree at the audited calibration. The recorded improvement over the ROM is a calibration-set result because per-output shrinkage was fitted on the same block; it is not counted as a fourth out-of-sample fact.

The corrected resource run establishes only that endpoint evaluation is possible under a tiny budget. It retained four draws in total, achieved 100 percent acceptance, and processed 0.0267 draws per second. Those numbers are not convergence evidence and are not a speed advantage. The acceptance rate instead suggests that the static-HMC tuning is far from an efficient production regime.

7 Claim-specific decision rules

Table 3 defines the minimum evidence needed to promote the working draft to a submission. Counts are protocol defaults, not universal publication standards; learning curves, uncertainty intervals, and reference convergence determine whether they are sufficient.

All thresholds and case identifiers will be frozen before the final runs. The evidence builder reads raw result files, verifies the full-model identity, hashes its sources, and emits both machine-readable and manuscript inputs. A failed hard gate remains visible in the replication package; it is not converted into a missing observation or silently omitted path.

8 Limitations and interpretation

The preserved evidence is an engineering pilot. Its nonlinear targets use branching order zero and one node, so it does not validate stochastic expectations. Most likelihood calls hold the reference steady state and ROM fixed. Its reported surrogate RMSE is calibration-adjusted rather than measured on an untouched test partition. The parameter-dependent full-JAX run still targets the surrogate profile inversion objective, assigns a finite -10^{12} sentinel to failed solves, and uses a host-callback Schur solve. An exact structural posterior instead requires explicit support handling and an economically defined likelihood. The sampler budgets are too small for convergence diagnostics,

Table 3: Hard gates for a credible reduced-model methods submission

Evidence family	Required result
Scope and provenance	Full 23-equation source verified by content hash and upstream revision; clean repository revision; complete environment, data, command, seed, and artifact manifests.
Nonlinear solution	Positive-order stochastic SEP with multiple nodes and all shocks active; raw attempted and successful counts; scaled residuals; regular, posterior, and boundary-stress strata.
Independent parity	Common case identifiers evaluated in Python and MacroModelling.jl; both failures retained in the denominator; QZ, trajectory, and likelihood errors reported on common successful cases.
Surrogate accuracy	Disjoint parameter, shock, and trajectory partitions; shrinkage disabled or fitted on a separate calibration partition; untouched-test per-output and tail errors; paired improvement over ROM with a 95 percent interval; learning-curve and out-of-distribution results.
Posterior fidelity	Two-parameter full nonlinear grid with resolution-doubling convergence; posterior moments, intervals, and distribution compared with the surrogate and corrected samplers using numerical or Monte Carlo uncertainty.
Sampling	Four tuned chains for the scaling experiment; rank-normalized split $\widehat{R}$, bulk and tail ESS, failures, ESS per gradient evaluation, and ESS per second.
Timing	Same-work, same-precision comparison; synchronized warm calls and independent end-to-end processes; compilation, transfers, training, host callbacks, and peak memory reported separately.

Notes: These are claim-specific design requirements. Prespecified thresholds and case lists must be frozen before confirmatory execution. Passing them supports the reduced Galí experiment only; it does not establish universal accuracy across DSGE models.

the data are synthetic and short, and no independent Julia trajectory or likelihood comparison is yet archived. The current evidence therefore supports neither exact nonlinear posterior sampling nor an end-to-end speedup claim.

These limitations determine the paper's next experiments rather than merely qualifying its conclusion. A credible methods result requires the two-parameter reference experiment to pass before scaling the posterior dimension. If the surrogate does not preserve the grid posterior, or if its full accounting does not improve ESS per second, the negative result should be reported and the claim reduced to numerical infrastructure.

9 Conclusion

Residual learning may be useful for nonlinear DSGE estimation because a first-order model provides a disciplined baseline and the network concentrates on the nonlinear discrepancy. Target-corrected HMC can remove error from the trajectory force, but only relative to the density actually evaluated at the Metropolis endpoints. The preserved Galí pilot shows that the software path is feasible and reveals where apparently favorable diagnostics can overstate the evidence. Publication readiness depends on stochastic-SEP validation, an untouched test design, an independently converged nonlinear posterior reference, cross-language parity, efficient multi-chain sampling, and same-work timing. The accompanying protocol makes those requirements falsifiable and preserves negative results rather than redefining success after a run.

Disclosure on computational assistance

Automated coding and language tools were used to inspect software, generate tests, and edit the working draft. Every numerical claim in the paper is linked to a machine-readable source file and must be verified by the authors. The final submission will follow the target journal's disclosure policy.

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